A factorial supersingular congruence and signed discriminant ratios

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Abstract

For every prime $p \geq 5$, we prove the supersingular congruence

$$E_{p+1}^{(p-1)/2} \equiv \left(\frac{6}{p}\right)\left(\frac{p-1}{2}\right)!\Delta^{(p^2-1)/24} \pmod{(p, E_{p-1})}.$$

It evaluates a half-power ratio uniformly on the supersingular locus and recovers Baker's congruence upon squaring. The proof applies an iterated theta operator to η^{p+1} and uses Euler's pentagonal identity to determine the sign. Combined with Robert's isogeny identity, it gives

$$\left(\frac{\Delta(\tau)}{\Delta(\ell\tau)}\right)^{(p^2-1)/24} = \left(\frac{\ell}{p}\right)$$

at every supersingular point of $X_0(\ell)$ for every prime $\ell \neq p$. This determines the half-exponent sign in Ono's supersingular discriminant-ratio identity and extends to arbitrary prime-to-p isogenies. A companion paper applies the identity to the residues of $T_N|U_p$ modulo p for Fricke Hauptmoduln T_N.

Keywords: supersingular congruences; Eisenstein series; discriminant ratios; isogenies; theta operator.

1 Introduction

Baker's supersingular congruence determines the square of a ratio formed from E_{p+1} and Δ (Baker, 1998). We evaluate the ratio itself explicitly, obtaining a factorial and a Legendre symbol that are independent of the supersingular elliptic curve. This refines Baker's congruence and, together with Robert's isogeny identity, determines the half-exponent sign of a normalized discriminant ratio along every prime-to-p isogeny. Neither result requires a genus-zero hypothesis.

A companion paper (Jordan, 2026) uses Theorem 1.3 below to determine the signed residues of the reduction of $T_N|U_p$ modulo p, where T_N is the Hauptmodul of a genus-zero Fricke quotient $X_0(N)^+$ of prime level.

1.1 Notation

Let $p \geq 5$ be prime and put

$$h = (p-1)/2, \qquad k = (p^2-1)/24, \qquad A = \overline{E}_{p-1}, \qquad B = \overline{E}_{p+1}, \qquad \Delta = \eta^{24}.$$

Here E_w is the normalized Eisenstein series of weight w, a bar denotes reduction modulo p, and A is the Hasse invariant. The q-expansions of A and B are 1 and E_2 modulo p, respectively. We write $q = e^{2\pi i\tau}$, $\theta = q\,d/dq$, $\mathbb{F}_p = \mathbb{F}_p$, $\mathbb{F}_{p^2} = \mathbb{F}_{p^2}$ and $\overline{\mathbb{F}}_p = \overline{\mathbb{F}}_p$. The symbol $\left(\frac{a}{p}\right)$ denotes the Legendre symbol. When a modular form is evaluated at an elliptic curve, a nonzero invariant differential is understood. Ratios of equal weight are independent of its choice. In particular, set

$$c(E) := \frac{B(E,\omega)^h}{\Delta(E,\omega)^k}.$$

1.2 The factorial congruence and discriminant ratios

The first result evaluates $c(E)$, rather than only its square.

Theorem 1.1 (supersingular factorial evaluation)**.** *For every prime $p \geq 5$, every supersingular elliptic curve $E/\overline{\mathbb{F}}_p$ and every nonzero invariant differential ω,*

$$B(E,\omega)^h = \left(\frac{6}{p}\right) h!\,\Delta(E,\omega)^k. \tag{1.1}$$

Equivalently, in the graded ring $\mathbb{F}_p[E_4, E_6]$,

$$A \mid B^h - \left(\frac{6}{p}\right) h!\,\Delta^k. \tag{1.2}$$

The proof in Section 2 applies the Serre–Robert theta-iteration method to $F = \eta^{p+1}$. Euler's pentagonal identity makes the relevant quadratic twist explicit, and comparison with the iterated operator determines the factorial and its sign uniformly in p. Squaring the evaluation recovers Baker's congruence.

Corollary 1.2 (constancy and Baker's congruence)**.** *The value $c(E)$ is the same for all supersingular $E/\overline{\mathbb{F}}_p$, namely $c_p = \left(\frac{6}{p}\right) h! \in \mathbb{F}_p^\times$. In particular $c_p^2 = -\left(\frac{-1}{p}\right)$, and*

$$E_{p+1}^{p-1} \equiv -\left(\frac{-1}{p}\right) \Delta^{(p^2-1)/12} \pmod{(p, E_{p-1})}.$$

For a prime ℓ, let $Y_0(\ell)$ denote the non-cuspidal modular curve over $\mathbb{Z}[1/\ell]$. For $x = (E, C)$, let $\pi : E \to E' = E/C$ and choose a nonzero invariant differential ω' on E'. Define

$$D_\ell(E, C) := \ell^{12} \frac{\Delta(E, \pi^*\omega')}{\Delta(E', \omega')}. \tag{1.3}$$

The differential weights make this independent of ω' and of the representative of x. It is a unit on $Y_0(\ell)$ over $\mathbb{Z}[1/\ell]$, with q-expansion $\Delta(q)/\Delta(q^\ell)$: on the Tate curve with $C = \mu_\ell$, $\pi^*\omega'_{\mathrm{can}} = \ell\omega_{\mathrm{can}}$. Its complex realization is therefore $\Delta(\tau)/\Delta(\ell\tau)$. We use the same notation for its reduction at $p \neq \ell$.

Theorem 1.3 (signed discriminant ratio)**.** *Let $p \geq 5$ and $\ell \neq p$ be primes. At every supersingular point x of $Y_0(\ell)_{\overline{\mathbb{F}}_p}$,*

$$D_\ell(x)^{(p^2-1)/24} = \left(\frac{\ell}{p}\right).$$

Ono (2026, Theorem 1.2) proves the corresponding identity at exponent $2k = (p^2-1)/12$ for the reciprocal ratio $\Delta(\ell\tau)/\Delta(\tau)$, and his Remark 4 exhibits values with nontrivial half-exponent sign. Theorem 1.3 determines that sign uniformly as $\left(\frac{\ell}{p}\right)$. Section 3 deduces the theorem from Theorem 1.1 and Robert's isogeny identity, and extends it to arbitrary prime-to-p isogenies. This is a refinement of the supersingular identity used by Ono; no new assertion about the Adams–Novikov spectral sequence is made here.

1.3 Relation to earlier work

Baker (1998) proves the supersingular congruence for B^{p-1} recovered in Corollary 1.2. Theorem 1.1 refines it by evaluating the half-power ratio $c(E)$ explicitly and showing that this value is independent of the supersingular curve. The isogeny transformation of B is due to Robert (1980, Théorème B, p. 106; §3.3, Lemma 7 and (19), p. 124); we include a q-expansion proof to fix the differential normalization.

The factorial evaluation uses a classical mechanism: Serre's functional equation for Eisenstein series, presented by Robert (1980, §2.5, Théorème 2, p. 119), who attributes the relation to a letter of Serre dated 13 June 1978 (Robert, 1980, §0.3, p. 109), produces factorial coefficients by iterating theta and then evaluating on the supersingular locus. We apply that mechanism to the form η^{p+1} of weight $(p+1)/2$ at full level twelve. Its explicit quadratic twist gives the signed value of c_p simultaneously in all residue classes of p modulo twelve. Robert's cited theorem concerns pairs of even-weight Eisenstein series with complementary weights summing to $p+1$; it does not state the factorial evaluation of Theorem 1.1. The argument here does not use Baker's congruence or the signed discriminant-ratio identity as an input.

Landweber (1988, Theorem 3, p. 72; §4) gives the half-exponent identity $\Delta^{(p^2-1)/24} = (e_a - e_b)^{(p^2-1)/4}$ for supersingular models $y^2 = 4\prod_{a=1}^{3}(x-e_a)$ and $a \neq b$. Theorem 1.3 instead compares discriminants along a prime-degree isogeny, with its differential normalization explicit, and determines the sign from Theorem 1.1 and Robert's identity. Ono (2026, Theorem 1.2 and Remark 4) proves the squared isogeny-ratio identity and records examples with nontrivial half-exponent sign. His work develops the supersingular congruences arising in the description of the divided beta family by Behrens (2009, Theorem 1.2) and the explicit calculations of Larson (2020, §5.4, Conjecture 5.17). These connections motivate the discriminant identity; no new statement about the Adams–Novikov spectral sequence is needed here.

Organization. Section 2 proves the factorial evaluation and Corollary 1.2. Section 3 proves the signed discriminant-ratio identity and its extension to arbitrary prime-to-p isogenies. Section 4 records exact-arithmetic checks.

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2 The supersingular factorial evaluation

Put $h = (p-1)/2$ and $m = (p+1)/2$, so $m+h = p$ and $m+h(p+1) = pm$. We prove Theorem 1.1 by comparing two expressions for an iterated theta operator applied to $F = \eta^{p+1}$. The leading coefficient calculation is the one underlying Serre's functional equation for Eisenstein series, as presented by Robert (1980, §2.5, Théorème 2, p. 119). The choice of F makes its quadratic twist explicit at every $p \geq 5$.

2.1 A form at full level twelve

Work on a geometrically connected component $X(12)$ of the compactified full-level modular curve, over $R = \mathbb{Z}_{(p)}[\zeta_{12}]$ localized at a prime above p. This is a smooth proper curve carrying the Hodge line bundle ω. Forms of weight w are sections of $\omega^{\otimes w}$; a choice of nonzero invariant differential on an elliptic curve allows evaluation of such a section. The cusp parameter is $q^{1/12}$. We use the integral q-expansion principle on this prime-to-p level (Katz, 1973, §1.6); see also Deligne and Rapoport, 1973.

Lemma 2.1. *The function η^2 is a holomorphic modular form of weight one with trivial character on $\Gamma(12)$. In particular $F = \eta^{p+1} \in M_m(\Gamma(12))$ is integral over R, and*

$$F^{24} = \Delta^{p+1}, \qquad F^{p-1} = \Delta^k. \tag{2.1}$$

Proof. The eta transformation law (Knopp, 1970, Ch. 4) gives $\eta^2(\gamma\tau) = \nu(\gamma)^2(c\tau+d)\eta^2(\tau)$. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ with $c > 0$ even, and hence d odd, the multiplier is

$$\nu(\gamma) = \left(\frac{c}{|d|}\right) \exp\left(\frac{\pi i}{12}\big((a+d)c - bd(c^2-1) + 3d - 3 - 3cd\big)\right),$$

where the first factor is a Jacobi symbol. This case applies to every $\gamma \in \Gamma(12)$ with $c > 0$, since $12 \mid c$. The symbol squares to one. Since $a \equiv d \equiv 1$ and $b \equiv c \equiv 0 \pmod{12}$, each of $(a+d)c$, $bd(c^2-1)$, $3(d-1)$ and $3cd$ is divisible by 12, so

$$\nu(\gamma)^2 = \exp\left(\frac{2\pi i}{12}\big((a+d)c - bd(c^2-1) + 3d - 3 - 3cd\big)\right) = 1.$$

For $c < 0$, apply the same formula to $-\gamma$: its numerator is -6 modulo 12, so its multiplier squared is -1; the factor $-c\tau - d$ supplies the other minus sign. For $c = 0$, γ is a translation by a multiple of 12 and the assertion follows from $\eta^2(\tau+12) = \eta^2(\tau)$. The form is holomorphic at every cusp. The transformation law and the product expansion give integral cusp expansions over R (the multipliers are twelfth roots of unity), so the q-expansion principle supplies the integral section. Raising to the mth power gives F. Both equalities in (2.1) follow from $\eta^{24} = \Delta$ and $24k = p^2 - 1$. □

2.2 The iterated operator and the twist

Let $\partial_w = \theta - \frac{w}{12}E_2$ be the Serre derivative. By the quasi-modular transformation law of E_2, it sends modular forms of weight w on $\Gamma(12)$ to modular forms of weight $w+2$ on $\Gamma(12)$. It also preserves p-integrality. Indeed, if f has expansion $\sum_n a_n q^{n/12}$ with all $a_n \in R$, then $\theta f = \sum_n (n/12)a_n q^{n/12}$ and $E_2 f$ again have coefficients in R, because 12 is a unit in R; the q-expansion principle (Katz, 1973, §1.6) then shows that $\partial_w f$ is defined over R. Define

$$G_0 = F, \qquad G_{i+1} = E_{p-1}\partial_{w_i} G_i + \frac{w_i}{12} E_{p+1} G_i, \qquad w_i = m + i(p+1).$$

Then G_i is a p-integral form of weight w_i. Modulo p, one step of this recursion is the operator $A\theta$ of Katz (1977), which raises the weight by $p+1$ and acts as θ on q-expansions. The Kummer congruences $E_{p-1} \equiv 1$ and $E_{p+1} \equiv E_2 \pmod{p}$ show that its q-expansion is $\theta^i F \pmod{p}$. Moreover

$$G_i = \gamma_i E_{p+1}^i F + E_{p-1} R_i, \qquad \gamma_i = 12^{-i} \prod_{j=0}^{i-1} w_j, \tag{2.2}$$

with R_i a p-integral modular form. This follows by induction: in the recursion the derivative term is multiplied by E_{p-1}, and the remaining term multiplies the leading coefficient by $w_i/12$. At $i = h$, Wilson's theorem gives

$$\overline{\gamma}_h = 12^{-h} \prod_{j=0}^{h-1} (m+j) = 12^{-h} \frac{(p-1)!}{h!} = -\frac{12^{-h}}{h!} \quad \text{in } \mathbb{F}_p. \tag{2.3}$$

Write $\chi_{12}(n) = \left(\frac{12}{n}\right)$ for the Kronecker character, with values $1, -1, 0$ on $n \equiv \pm 1, \pm 5$, and nonunits modulo 12, respectively. Euler's pentagonal identity (Apostol, 1990) takes the form

$$\eta(\tau) = \sum_{n \geq 1} \chi_{12}(n) q^{n^2/24}.$$

Lemma 2.2 (quadratic twist)**.** *The following identity holds between mod-p forms of weight pm on $\Gamma(12)$:*

$$\overline{G}_h = \left(\frac{6}{p}\right)(A^m \overline{F} - \chi_{12}(p)\overline{F}^p). \tag{2.4}$$

Proof. By the product expansion and the freshman's dream, $F \equiv \eta(\tau)\eta(p\tau) \pmod p$. Thus

$$\overline{F} = \sum_{a,b\geq 1} \chi_{12}(a)\chi_{12}(b) q^{(a^2+pb^2)/24}.$$

Applying θ^h multiplies each term by $((a^2+pb^2)/24)^h$, which is zero modulo p when $p \mid a$ and equals $\left(\frac{24}{p}\right) = \left(\frac{6}{p}\right)$ otherwise. The sub-sum with $a = pa'$ is

$$\chi_{12}(p) \sum_{a',b\geq 1} \chi_{12}(a')\chi_{12}(b) q^{p(b^2+pa'^2)/24} = \chi_{12}(p)\overline{F}^p.$$

Consequently the q-expansion of $\overline{G}_h$ equals $\left(\frac{6}{p}\right)(\overline{F} - \chi_{12}(p)\overline{F}^p)$. The integral forms G_h, $E_{p-1}^m F$ and F^p all have weight pm. Since A has q-expansion one, the q-expansion principle proves (2.4) as an identity of sections. □

Proof of Theorem 1.1. Let (E, ω_E) be supersingular and choose a full level-12 structure α with the fixed Weil pairing. Evaluate (2.2) and (2.4) at $x = (E, \alpha, \omega_E)$. Since $A(x) = 0$, they give

$$-\frac{12^{-h}}{h!} B(E,\omega_E)^h \overline{F}(x) = -\left(\frac{6}{p}\right) \chi_{12}(p)\overline{F}(x)^p.$$

Equation (2.1) shows that $\overline{F}(x) \neq 0$ and $\overline{F}(x)^{p-1} = \Delta(E,\omega_E)^k$. Hence

$$B(E,\omega_E)^h = \left(\frac{6}{p}\right) \chi_{12}(p) 12^h h!\, \Delta(E,\omega_E)^k.$$

Euler's criterion gives $12^h = \left(\frac{12}{p}\right) = \chi_{12}(p)$, so the two character factors cancel. This proves the pointwise formula for every supersingular curve, including non-rational j-invariants.

For the divisibility statement, work in $\overline{\mathbb{F}}_p[E_4, E_6]$. The Hasse invariant A is squarefree and prime to B (Robert, 1980, §2.3, Théorème 1(ii), p. 116). Its factors are E_4 if $j = 0$ is supersingular, E_6 if $j = 1728$ is supersingular, and $E_4^3 - j_0\Delta$ for the remaining supersingular j_0 (Kaneko and Zagier, 1998, §3). This is the geometric factorization over $\overline{\mathbb{F}}_p$; the irreducible quadratic factors in j over $\mathbb{F}_p$ in Baker (1998, Theorem 1.2) split after extending the constants. For $j_0 \neq 0, 1728$, the factor

$$E_4^3 - j_0\Delta = \left(1 - \frac{j_0}{1728}\right) E_4^3 + \frac{j_0}{1728} E_6^2$$

is irreducible: it has the form $aX^3 + bY^2$ with $ab \neq 0$. The zero set of each factor away from the origin parametrizes the corresponding curves with their nonzero differentials. The pointwise identity therefore forces each factor to divide $B^h - \left(\frac{6}{p}\right) h!\Delta^k$, so their product A divides it. Divisibility descends to $\mathbb{F}_p[E_4, E_6]$. Conversely, divisibility immediately implies the pointwise identity because A vanishes at every supersingular curve. The factors E_4 and E_6 account for the elliptic points $j = 0, 1728$ without extra multiplicities in the graded ring. □

Proof of Corollary 1.2. Theorem 1.1 gives $c(E) = \left(\frac{6}{p}\right) h! \in \mathbb{F}_p^\times$ independently of E. Squaring and using $(h!)^2 = (-1)^{h+1} = -\left(\frac{-1}{p}\right)$ proves the stated form of Baker's congruence. This proof uses neither that congruence nor connectivity of supersingular isogeny graphs as an input. □

Remark 2.3 (the constant c_p). The constant $c_p = \left(\frac{6}{p}\right) h!$ is not an elementary sign. If $p \equiv 1 \pmod 4$, then $c_p^2 = -1$, so c_p is a square root of -1 in $\mathbb{F}_p$. If $p \equiv 3 \pmod 4$, then $c_p = \pm 1$, and the sign is governed by a class number: writing $\mathrm{cl}(-p)$ for the class number of $\mathbb{Q}(\sqrt{-p})$, Mordell's congruence $h! \equiv (-1)^{(\mathrm{cl}(-p)+1)/2} \pmod p$ (Mordell, 1961) gives

$$c_p = \left(\frac{6}{p}\right) (-1)^{(\mathrm{cl}(-p)+1)/2} \qquad (p \equiv 3 \pmod 4).$$

For example, $c_7 = c_{11} = c_{23} = 1$ and $c_{19} = c_{31} = -1$.

Remark 2.4 (a limit to higher-root constancy). Constancy in Corollary 1.2 need not persist for a further square root. For $p \equiv 1 \pmod 8$, the ratio $\rho_p = B^{(p-1)/4}/\Delta^{(p^2-1)/48}$ has weight zero and satisfies $\rho_p^2 = c_p$ at supersingular points, but it need not be constant. For example, modulo 17,

$$A = 8E_4^4 + 10E_4E_6^2, \qquad B = 7E_4^3E_6 + 11E_6^3.$$

The two pairs of form values below have $A = 0$ and $\Delta \neq 0$, and hence represent supersingular elliptic curves with nonzero invariant differentials. Using $\Delta = (E_4^3 - E_6^2)/1728$ gives

j	E_4	E_6	B	Δ	$\rho_{17} = B^4/\Delta^6$
0	0	1	11	3	-2
8	3	3	14	14	2

in $\mathbb{F}_{17}$.

Both values square to $c_{17} = 4$. Thus a uniformly constant quarter-power analogue is false. More generally, whenever $r \mid h$ and $r \mid k$, the ratio $B^{h/r}/\Delta^{k/r}$ takes at most r values, since its rth power is c_p; this bound alone does not imply that all r values occur.

3 Signed discriminant ratios

The factorial evaluation makes $c(E)$ constant on the supersingular locus. Robert's isogeny identity compares B on the two curves of an isogeny and turns this constancy into a formula for their discriminants. We first record the differential normalization.

Lemma 3.1 (Robert's isogeny identity)**.** *Let ℓ be any prime $\neq p$. Let $x = (E, C)$ be a supersingular point of $X_0(\ell)$, $\pi : E \to E' = E/C$, and $\omega = \pi^*\omega'$. Then $B(E', \omega') = \ell\, B(E, \omega)$, $A(E', \omega') = A(E, \omega)$ and $\Delta(\ell\tau)(x) = \ell^{-12}\Delta(E', \omega')$.*

Proof. The isogeny identity is Robert (1980, Théorème B; §3.3, Lemma 7); see also Baker (1998, Lemma 2.2). We verify the normalization by q-expansions. For a level-one form f of weight w, $f(\ell\tau)$ at (E, C, ω) equals $\ell^{-w} f(E', \omega')$. Check this on $(\mathrm{Tate}(q), \mu_\ell)$, where $E' \cong \mathrm{Tate}(q^\ell)$ and $\pi^*\omega'_{\mathrm{can}} = \ell\omega_{\mathrm{can}}$. The form $E_2(\tau) - \ell E_2(\ell\tau)$ is a holomorphic weight-2 form on $\Gamma_0(\ell)$ with p-integral coefficients. Modulo p, the forms $B(\tau) - \ell B(\ell\tau)$ and $A \cdot (E_2 - \ell E_2(\ell\tau))$ have weight $p + 1$ and the same q-expansion, so they are equal. At a supersingular point this gives $B(E, \omega) = \ell \cdot \ell^{-(p+1)} B(E', \omega') = \ell^{-1} B(E', \omega')$, because $\ell^{p-1} = 1$. The statement about A follows in the same way from $A(\ell\tau) = A(\tau)$. □

We use the algebraic unit D_ℓ of (1.3), with $\pi : E \to E' = E/C$ and $\omega = \pi^*\omega'$.

Proof of Theorem 1.3. At a supersingular point $x = (E, C)$, Lemma 3.1 and (1.3) imply

$$\frac{c(E')}{c(E)} = \ell^h \left(\frac{\Delta(E, \omega)}{\Delta(E', \omega')}\right)^k = \left(\frac{\ell}{p}\right) D_\ell(x)^k,$$

because $\ell^{12k} = \ell^{(p^2-1)/2} = 1$. By Corollary 1.2, the left side is one and is nonzero. Therefore $D_\ell(x)^k = \left(\frac{\ell}{p}\right)$. □

Corollary 3.2 (arbitrary prime-to-p isogenies)**.** *Let $\varphi : E \to E'$ be an isogeny of degree $n \geq 1$ between supersingular elliptic curves over $\overline{\mathbb{F}}_p$, with $(n, p) = 1$. For any nonzero invariant differential ω' on E', put*

$$D_\varphi := n^{12} \frac{\Delta(E, \varphi^*\omega')}{\Delta(E', \omega')}.$$

Then D_φ is independent of ω' and

$$D_\varphi^k = \left(\frac{n}{p}\right).$$

In particular, for cyclic kernels the signed discriminant-ratio identity holds on $Y_0(n)_{\overline{\mathbb{F}}_p}$ for every n prime to p.

Proof. Over $\overline{\mathbb{F}}_p$, factor φ into isogenies of prime degree and apply Lemma 3.1 successively, choosing differentials by pullback. This gives $B(E',\omega') = nB(E,\varphi^*\omega')$. Consequently

$$1 = \frac{c(E')}{c(E)} = n^h \left(\frac{\Delta(E,\varphi^*\omega')}{\Delta(E',\omega')}\right)^k = \left(\frac{n}{p}\right) D_\varphi^k,$$

since $n^{12k} = 1$ in $\mathbb{F}_p$. The Legendre symbol is its own inverse. The weight of Δ gives independence of the chosen differential. □

Remark 3.3 (eta coordinates). For $N \in \{2,3,5,7,13\}$ put $a_N = 24/(N-1)$ and $u_N = (\eta(\tau)/\eta(N\tau))^{a_N}$, a coordinate on $X_0(N)$. Then $D_N = u_N^{N-1}$, and Theorem 1.3 gives

$$u_N(x)^{(p^2-1)/a_N} = \left(\frac{N}{p}\right)$$

at every supersingular point x of $Y_0(N)_{\overline{\mathbb{F}}_p}$, for every prime $p \geq 5$ with $p \neq N$. For example, $j = (u_2+256)^3/u_2^2$ on $X_0(2)$. For $p = 5$ the only supersingular invariant is $j = 0$, attained at $u_2 = -256 \equiv -1 = \left(\frac{2}{5}\right)$. For $p = 7$ it is $j = 1728 \equiv 6$, attained exactly at $u_2 \equiv \pm 1$, and $u_2^2 = 1 = \left(\frac{2}{7}\right)$. This is the form in which the identity is used in the companion paper (Jordan, 2026).

Remark 3.4 (comparison with Ono). Ono (2026, Theorem 1.2 and Remark 4) uses the reciprocal $R_\ell = \Delta(\ell\tau)/\Delta(\tau) = D_\ell^{-1}$. Since a Legendre symbol equals its inverse, Theorem 1.3 determines the same half-exponent sign for R_ℓ, namely $\left(\frac{\ell}{p}\right)$. For example, $\left(\frac{2}{11}\right) = -1$, as in his example. His Theorem 1.3 states that on a model with p^2-Frobenius $[-p]$ and an $\mathbb{F}_{p^2}$-rational differential, $\Delta(E,\omega)^{2k} = (-1)^{(p+1)/2}$. Theorem 1.1 instead relates $B(E,\omega)^h$ to $\Delta(E,\omega)^k$ for every nonzero differential. The ratio argument requires no descent of a twenty-fourth root of Δ to $\mathbb{F}_{p^2}$.

4 Computational evidence

The following exact-arithmetic computations are checks on the theorems, not inputs to their proofs.

For Theorem 1.1, exact polynomial remainders and evaluations at all 6430 supersingular j-values for the 166 primes $5 \leq p \leq 997$ (4734 outside $\mathbb{F}_p$) agree with the asserted constant. Evaluations use three randomly rescaled models per value, with additional Legendre and Weierstrass coefficient routes. The two theta identities in the proof were also checked in $\mathbb{F}_p[E_4, E_6]$ at $p = 23, 47, 71, 167$, where η^{p+1} has level one.

Theorem 1.3 was tested independently using normalized Vélu quotients and discriminants, without eta functions or Hauptmoduln. For the 15 primes $\ell \leq 47$ and admissible $5 \leq p \leq 199$, all 647 pairs and 127541 cyclic-kernel checks gave the asserted discriminant-ratio sign. These count kernels on chosen supersingular representatives, not distinct coarse points, since automorphisms can identify kernels at $j = 0, 1728$.

Data and software. The supplementary material for this paper consists of three packages. Their folder names are those of the archived release; they do not reflect the current status or numbering of the results.

- `Factorial_Conjecture_Result_v1_0`: code and recorded results for Theorem 1.1 under `new_work/`, with reproduction commands in `run_all.sh` (`cypari2` 2.2.4, PARI/GP 2.17.2 and `mpmath` 1.3.0).
- `Eta_Frobenius_Lemma_Audit_v1_0`: the Vélu-quotient checks of Theorem 1.3.

- `Theorem_1_3_Benchmark_v1_0`: timings for exact evaluation of B^h by Theorem 1.1 (numbered Theorem 1.3 in an earlier draft), with the timing code, exact inputs and outputs, raw timing samples and verification records.

The code, data and reproduction instructions are publicly available at https://github.com/bbrhuft/factorial-supersingular-congruence/tree/main/supplementary.

5 Further directions

The proof of Theorem 1.1 suggests seeking other families of modular forms whose quadratic twists admit similarly explicit expressions. Extending this proof strategy would require a suitable analogue of the twist identity, together with the necessary weight, integrality and nonvanishing properties. Remark 2.4 shows that further root extraction need not preserve constancy: the half-power evaluation cannot in general be replaced by a uniformly constant quarter-power analogue.

The discriminant-ratio identity holds without a genus-zero hypothesis, including for arbitrary prime-to-p isogenies. Its application to Fricke Hauptmoduln, which additionally requires an integral genus-zero coordinate, is the subject of the companion paper (Jordan, 2026).

Code and data availability

All code, scripts and logs are available in the GitHub repository

https://github.com/bbrhuft/factorial-supersingular-congruence

The computational materials described here are those of release v2.0 of this repository. Earlier versions of this work, in which this paper and its companion formed a single manuscript, are archived on Zenodo at doi:10.5281/zenodo.24024359 (v1.11) and doi:10.5281/zenodo.24084789 (v1.13).

References

Apostol, T. M. (1990). *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed. Graduate Texts in Mathematics 41. Springer.

Baker, A. (1998). A supersingular congruence for modular forms. *Acta Arithmetica*, 86(1), 91–100. https://doi.org/10.4064/aa-86-1-91-100.

Behrens, M. (2009). Congruences between modular forms given by the divided β family in homotopy theory. *Geometry & Topology*, 13, 319–357. https://doi.org/10.2140/gt.2009.13.319.

Deligne, P., & Rapoport, M. (1973). Les schémas de modules de courbes elliptiques. In *Modular Functions of One Variable II* (LNM 349, pp. 143–316). Springer.

Jordan, D. (2026). Supersingular poles and signed residues of prime-level Fricke Hauptmoduln. Preprint (companion paper). Manuscript, code and data: https://github.com/bbrhuft/fricke-signed-residues.

Kaneko, M., & Zagier, D. (1998). Supersingular j-invariants, hypergeometric series, and Atkin's orthogonal polynomials. In *Computational Perspectives on Number Theory* (AMS/IP Studies in Advanced Mathematics 7, pp. 97–126).

Katz, N. M. (1973). p-adic properties of modular schemes and modular forms. In *Modular Functions of One Variable III* (LNM 350, pp. 69–190). Springer.

Katz, N. M. (1977). A result on modular forms in characteristic p. In *Modular Functions of One Variable V* (LNM 601, pp. 53–61). Springer.

Knopp, M. I. (1970). *Modular Functions in Analytic Number Theory*. Markham.

Landweber, P. S. (1988). Supersingular elliptic curves and congruences for Legendre polynomials. In *Elliptic Curves and Modular Forms in Algebraic Topology*, Lecture Notes in Mathematics 1326, pp. 69–93. Springer.

Larson, D. M. (2020). Explicit modular forms from the divided beta family. *Topology and its Applications*, 273, 107094. https://doi.org/10.1016/j.topol.2020.107094.

Mordell, L. J. (1961). The congruence $((p-1)/2)! \equiv \pm 1 \pmod{p}$. *American Mathematical Monthly*, 68, 145–146.

Ono, K. (2026). Modular forms for chromatic homotopy: supersingular congruences. *arXiv:2509.16175v2*, 3 July 2026. https://arxiv.org/abs/2509.16175v2.

Robert, G. (1980). Congruences entre séries d'Eisenstein, dans le cas supersingulier. *Inventiones Mathematicae*, 61, 103–158.

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